For Blocks: If the calculated F_{0b} value is greater than the corresponding critical value, then we reject the null hypothesis and conclude that there is significant difference among the block means, in at least one pair.

3.4.2 Merits and Demerits of two-way ANOVA

Merits

- Any number of blocks and treatments can be used.
- Number of units in each block should be equal.
- It is the most used design in view of the smaller total sample size since we are studying two variable at a time.

Demerits

- If the number of treatments is large enough, then it becomes difficult to maintain the homogeneity of the blocks.
- If there is a missing value, it cannot be ignored. It has to be replaced with some function of the existing values and certain adjustments have to be made in the analysis. This makes the analysis slightly complex.

Comparison between one-way ANOVA and two-way ANOVA

Basis of comparison	ANOVA	
	One-way	Two-way
Independent variable	One	Two
Compares	Three or more levels of one factor	Three or more levels of two factors, simultaneously
Number of observations	Need not be same in each treatment group	Need to be equal in each treatment group

Example 3.6

A reputed marketing agency in India has three different training programs for its salesmen. The three programs are Method – A, B, C. To assess the success of the programs, 4 salesmen from each of the programs were sent to the field. Their performances in terms of sales are given in the following table.

Salesmen	Methods		
	A	B	C
1	4	6	2
2	6	10	6
3	5	7	4
4	7	5	4

Test whether there is significant difference among methods and among salesmen.

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